MongoDB script for inserting:

```
use ecommerce
// Insert product_details with reviews
for (let i = 1; i <= 20; i++) {
    db.product_details.insertOne({
        product_id: i,
        description: "Product description " + i,
        images: ["img" + i + "_1.jpg", "img" + i + "_2.jpg"],
        attributes: {
            brand: "Brand" + i,
            color: "Color" + (i % 5),
            warranty: "1 year"
        },
        reviews: [
            {
                user_id: Math.floor(Math.random()*20)+1,
                rating: Math.floor(Math.random()*5)+1,
                comment: "Sample review for product " + i
            }
        ]
    })
}

// Insert user activity logs
for (let i = 1; i <= 20; i++) {
    db.user_activity.insertOne({
        user_id: i,
        actions: [
            {
                type: "view_product",
                product_id: Math.floor(Math.random()*50)+1,
                timestamp: new Date(),
```

```
                details: "Viewed product page"
            },
            {
                type: "add_to_cart",
                product_id: Math.floor(Math.random()*50)+1,
                timestamp: new Date(),
                details: "Added product to cart"
            }
        ]
    })
}
```

The screenshots for the populated databases are shown next page onwards:

`db.product_details.find().pretty()`

displays Product_details content

```
[
  {
    _id: ObjectId('69aa84576c96fe95a28563b1'),
    product_id: 1,
    description: 'High performance gaming laptop',
    images: [ 'img1.jpg', 'img2.jpg' ],
    attributes: { brand: 'ASUS', ram: '16GB', storage: '1TB SSD' },
    reviews: [
      { user: 'Amit', rating: 5, comment: 'Excellent performance' },
      { user: 'Priya', rating: 4, comment: 'Good but slightly expensive' }
    ]
  },
  {
    _id: ObjectId('69b40d0d5fc2c40c588563b1'),
    product_id: 1,
    description: 'Product description 1',
    images: [ 'img1_1.jpg', 'img1_2.jpg' ],
    attributes: { brand: 'Brand1', color: 'Color1', warranty: '1 year' },
    reviews: [ { user_id: 15, rating: 3, comment: 'Sample review for product 1' } ]
  },
  {
    _id: ObjectId('69b40d0d5fc2c40c588563b2'),
    product_id: 2,
    description: 'Product description 2',
    images: [ 'img2_1.jpg', 'img2_2.jpg' ],
    attributes: { brand: 'Brand2', color: 'Color2', warranty: '1 year' },
    reviews: [ { user_id: 10, rating: 5, comment: 'Sample review for product 2' } ]
  },
  {
    _id: ObjectId('69b40d0d5fc2c40c588563b3'),
    product_id: 3,
    description: 'Product description 3',
    images: [ 'img3_1.jpg', 'img3_2.jpg' ],
    attributes: { brand: 'Brand3', color: 'Color3', warranty: '1 year' },
    reviews: [ { user_id: 19, rating: 3, comment: 'Sample review for product 3' } ]
  },
  {
    _id: ObjectId('69b40d0d5fc2c40c588563b4'),
    product_id: 4,
    description: 'Product description 4',
    images: [ 'img4_1.jpg', 'img4_2.jpg' ],
    attributes: { brand: 'Brand4', color: 'Color4', warranty: '1 year' },
    reviews: [ { user_id: 14, rating: 2, comment: 'Sample review for product 4' } ]
  },
  {
    _id: ObjectId('69b40d0d5fc2c40c588563b5'),
    product_id: 5,
    description: 'Product description 5',
    images: [ 'img5_1.jpg', 'img5_2.jpg' ],
    attributes: { brand: 'Brand5', color: 'Color0', warranty: '1 year' },
    reviews: [ { user_id: 10, rating: 5, comment: 'Sample review for product 5' } ]
  },
  {
    _id: ObjectId('69b40d0d5fc2c40c588563b6'),
    product_id: 6,
    description: 'Product description 6',
    images: [ 'img6_1.jpg', 'img6_2.jpg' ],
    attributes: { brand: 'Brand6', color: 'Color1', warranty: '1 year' },
    reviews: [ { user_id: 3, rating: 3, comment: 'Sample review for product 6' } ]
  },
  {
    _id: ObjectId('69b40d0d5fc2c40c588563b7'),
    product_id: 7,
    description: 'Product description 7',
    images: [ 'img7_1.jpg', 'img7_2.jpg' ],
    attributes: { brand: 'Brand7', color: 'Color2', warranty: '1 year' },
    reviews: [ { user_id: 16, rating: 1, comment: 'Sample review for product 7' } ]
  },
  {
    _id: ObjectId('69b40d0d5fc2c40c588563b8'),
    product_id: 8,
    description: 'Product description 8',
    images: [ 'img8_1.jpg', 'img8_2.jpg' ],
    attributes: { brand: 'Brand8', color: 'Color3', warranty: '1 year' },
    reviews: [ { user_id: 13, rating: 4, comment: 'Sample review for product 8' } ]
  },
  {
    _id: ObjectId('69b40d0d5fc2c40c588563b9'),
    product_id: 9,
```

```
  description: 'Product description 9',
  images: [ 'img9_1.jpg', 'img9_2.jpg' ],
  attributes: { brand: 'Brand9', color: 'Color4', warranty: '1 year' },
  reviews: [ { user_id: 9, rating: 2, comment: 'Sample review for product 9' } ]
},
{
  _id: ObjectId('69b40d0d5fc2c40c588563ba'),
  product_id: 10,
  description: 'Product description 10',
  images: [ 'img10_1.jpg', 'img10_2.jpg' ],
  attributes: { brand: 'Brand10', color: 'Color0', warranty: '1 year' },
  reviews: [ { user_id: 15, rating: 5, comment: 'Sample review for product 10' } ]
},
{
  _id: ObjectId('69b40d0d5fc2c40c588563bb'),
  product_id: 11,
  description: 'Product description 11',
  images: [ 'img11_1.jpg', 'img11_2.jpg' ],
  attributes: { brand: 'Brand11', color: 'Color1', warranty: '1 year' },
  reviews: [ { user_id: 14, rating: 1, comment: 'Sample review for product 11' } ]
},
{
  _id: ObjectId('69b40d0d5fc2c40c588563bc'),
  product_id: 12,
  description: 'Product description 12',
  images: [ 'img12_1.jpg', 'img12_2.jpg' ],
  attributes: { brand: 'Brand12', color: 'Color2', warranty: '1 year' },
  reviews: [ { user_id: 20, rating: 2, comment: 'Sample review for product 12' } ]
},
{
  _id: ObjectId('69b40d0d5fc2c40c588563bd'),
  product_id: 13,
  description: 'Product description 13',
  images: [ 'img13_1.jpg', 'img13_2.jpg' ],
  attributes: { brand: 'Brand13', color: 'Color3', warranty: '1 year' },
  reviews: [ { user_id: 8, rating: 5, comment: 'Sample review for product 13' } ]
},
{
  _id: ObjectId('69b40d0d5fc2c40c588563be'),
  product_id: 14,
```

```
  description: 'Product description 14',
  images: [ 'img14_1.jpg', 'img14_2.jpg' ],
  attributes: { brand: 'Brand14', color: 'Color4', warranty: '1 year' },
  reviews: [ { user_id: 13, rating: 3, comment: 'Sample review for product 14' } ]
},
{
  _id: ObjectId('69b40d0d5fc2c40c588563bf'),
  product_id: 15,
  description: 'Product description 15',
  images: [ 'img15_1.jpg', 'img15_2.jpg' ],
  attributes: { brand: 'Brand15', color: 'Color0', warranty: '1 year' },
  reviews: [ { user_id: 15, rating: 3, comment: 'Sample review for product 15' } ]
},
{
  _id: ObjectId('69b40d0d5fc2c40c588563c0'),
  product_id: 16,
  description: 'Product description 16',
  images: [ 'img16_1.jpg', 'img16_2.jpg' ],
  attributes: { brand: 'Brand16', color: 'Color1', warranty: '1 year' },
  reviews: [ { user_id: 10, rating: 4, comment: 'Sample review for product 16' } ]
},
{
  _id: ObjectId('69b40d0d5fc2c40c588563c1'),
  product_id: 17,
  description: 'Product description 17',
  images: [ 'img17_1.jpg', 'img17_2.jpg' ],
  attributes: { brand: 'Brand17', color: 'Color2', warranty: '1 year' },
  reviews: [ { user_id: 10, rating: 2, comment: 'Sample review for product 17' } ]
},
{
  _id: ObjectId('69b40d0d5fc2c40c588563c2'),
  product_id: 18,
  description: 'Product description 18',
  images: [ 'img18_1.jpg', 'img18_2.jpg' ],
  attributes: { brand: 'Brand18', color: 'Color3', warranty: '1 year' },
  reviews: [ { user_id: 2, rating: 1, comment: 'Sample review for product 18' } ]
},
{
  _id: ObjectId('69b40d0d5fc2c40c588563c3'),
  product_id: 19,
```

```
    description: 'Product description 19',
    images: [ 'img19_1.jpg', 'img19_2.jpg' ],
    attributes: { brand: 'Brand19', color: 'Color4', warranty: '1 year' },
    reviews: [ { user_id: 4, rating: 2, comment: 'Sample review for product 19' } ]
  }
]
Type "it" for more
ecommerce> it
[
  {
    _id: ObjectId('69b40d0d5fc2c40c588563c4'),
    product_id: 20,
    description: 'Product description 20',
    images: [ 'img20_1.jpg', 'img20_2.jpg' ],
    attributes: { brand: 'Brand20', color: 'Color0', warranty: '1 year' },
    reviews: [ { user_id: 15, rating: 4, comment: 'Sample review for product 20' } ]
  }
]
```

The screenshots for the populated user_activity database are shown next page onwards:

`db.user_details.find().pretty()`

displays User_details content.

```
{
  _id: ObjectId('69b40d0e5fc2c40c588563c5'),
  user_id: 1,
  actions: [
    {
      type: 'view_product',
      product_id: 30,
      timestamp: ISODate('2026-03-13T13:11:42.043Z'),
      details: 'Viewed product page'
    },
    {
      type: 'add_to_cart',
      product_id: 12,
      timestamp: ISODate('2026-03-13T13:11:42.043Z'),
      details: 'Added product to cart'
    }
  ]
},
{
  _id: ObjectId('69b40d0e5fc2c40c588563c6'),
  user_id: 2,
  actions: [
    {
      type: 'view_product',
      product_id: 12,
      timestamp: ISODate('2026-03-13T13:11:42.086Z'),
      details: 'Viewed product page'
    },
    {
      type: 'add_to_cart',
      product_id: 15,
      timestamp: ISODate('2026-03-13T13:11:42.086Z'),
      details: 'Added product to cart'
    }
  ]
},
{
  _id: ObjectId('69b40d0e5fc2c40c588563c7'),
  user_id: 3,
  actions: [
    {
      type: 'view_product',
      product_id: 1,
      timestamp: ISODate('2026-03-13T13:11:42.088Z'),
      details: 'Viewed product page'
    },
    {
      type: 'add_to_cart',
      product_id: 43,
      timestamp: ISODate('2026-03-13T13:11:42.088Z'),
      details: 'Added product to cart'
    }
  ]
},
{
  _id: ObjectId('69b40d0e5fc2c40c588563c8'),
  user_id: 4,
  actions: [
    {
      type: 'view_product',
      product_id: 16,
      timestamp: ISODate('2026-03-13T13:11:42.088Z'),
      details: 'Viewed product page'
    },
    {
      type: 'add_to_cart',
      product_id: 14,
      timestamp: ISODate('2026-03-13T13:11:42.088Z'),
      details: 'Added product to cart'
    }
  ]
},
{
  _id: ObjectId('69b40d0e5fc2c40c588563c9'),
  user_id: 5,
  actions: [
    {
      type: 'view_product',
      product_id: 48,
      timestamp: ISODate('2026-03-13T13:11:42.089Z'),
```

```
      details: 'Viewed product page'
    },
    {
      type: 'add_to_cart',
      product_id: 13,
      timestamp: ISODate('2026-03-13T13:11:42.089Z'),
      details: 'Added product to cart'
    }
  ]
},
{
  _id: ObjectId('69b40d0e5fc2c40c588563ca'),
  user_id: 6,
  actions: [
    {
      type: 'view_product',
      product_id: 25,
      timestamp: ISODate('2026-03-13T13:11:42.090Z'),
      details: 'Viewed product page'
    },
    {
      type: 'add_to_cart',
      product_id: 36,
      timestamp: ISODate('2026-03-13T13:11:42.090Z'),
      details: 'Added product to cart'
    }
  ]
},
{
  _id: ObjectId('69b40d0e5fc2c40c588563cb'),
  user_id: 7,
  actions: [
    {
      type: 'view_product',
      product_id: 28,
      timestamp: ISODate('2026-03-13T13:11:42.091Z'),
      details: 'Viewed product page'
    },
    {
      type: 'add_to_cart',
      product_id: 34,
      timestamp: ISODate('2026-03-13T13:11:42.091Z'),
      details: 'Added product to cart'
    }
  ]
},
{
  _id: ObjectId('69b40d0e5fc2c40c588563cc'),
  user_id: 8,
  actions: [
    {
      type: 'view_product',
      product_id: 2,
      timestamp: ISODate('2026-03-13T13:11:42.092Z'),
      details: 'Viewed product page'
    },
    {
      type: 'add_to_cart',
      product_id: 23,
      timestamp: ISODate('2026-03-13T13:11:42.092Z'),
      details: 'Added product to cart'
    }
  ]
},
{
  _id: ObjectId('69b40d0e5fc2c40c588563cd'),
  user_id: 9,
  actions: [
    {
      type: 'view_product',
      product_id: 5,
      timestamp: ISODate('2026-03-13T13:11:42.093Z'),
      details: 'Viewed product page'
    },
    {
      type: 'add_to_cart',
      product_id: 39,
      timestamp: ISODate('2026-03-13T13:11:42.093Z'),
      details: 'Added product to cart'
    }
```

```
    ]
  },
  {
    _id: ObjectId('69b40d0e5fc2c40c588563ce'),
    user_id: 10,
    actions: [
      {
        type: 'view_product',
        product_id: 5,
        timestamp: ISODate('2026-03-13T13:11:42.094Z'),
        details: 'Viewed product page'
      },
      {
        type: 'add_to_cart',
        product_id: 30,
        timestamp: ISODate('2026-03-13T13:11:42.094Z'),
        details: 'Added product to cart'
      }
    ]
  },
  {
    _id: ObjectId('69b40d0e5fc2c40c588563cf'),
    user_id: 11,
    actions: [
      {
        type: 'view_product',
        product_id: 30,
        timestamp: ISODate('2026-03-13T13:11:42.094Z'),
        details: 'Viewed product page'
      },
      {
        type: 'add_to_cart',
        product_id: 35,
        timestamp: ISODate('2026-03-13T13:11:42.094Z'),
        details: 'Added product to cart'
      }
    ]
  },
  {
    _id: ObjectId('69b40d0e5fc2c40c588563d0'),
```

```
    user_id: 12,
    actions: [
      {
        type: 'view_product',
        product_id: 28,
        timestamp: ISODate('2026-03-13T13:11:42.095Z'),
        details: 'Viewed product page'
      },
      {
        type: 'add_to_cart',
        product_id: 4,
        timestamp: ISODate('2026-03-13T13:11:42.095Z'),
        details: 'Added product to cart'
      }
    ]
  },
  {
    _id: ObjectId('69b40d0e5fc2c40c588563d1'),
    user_id: 13,
    actions: [
      {
        type: 'view_product',
        product_id: 20,
        timestamp: ISODate('2026-03-13T13:11:42.096Z'),
        details: 'Viewed product page'
      },
      {
        type: 'add_to_cart',
        product_id: 19,
        timestamp: ISODate('2026-03-13T13:11:42.096Z'),
        details: 'Added product to cart'
      }
    ]
  },
  {
    _id: ObjectId('69b40d0e5fc2c40c588563d2'),
    user_id: 14,
    actions: [
      {
        type: 'view_product',
```

```
      product_id: 39,
      timestamp: ISODate('2026-03-13T13:11:42.097Z'),
      details: 'Viewed product page'
    },
    {
      type: 'add_to_cart',
      product_id: 28,
      timestamp: ISODate('2026-03-13T13:11:42.097Z'),
      details: 'Added product to cart'
    }
  ]
},
{
  _id: ObjectId('69b40d0e5fc2c40c588563d3'),
  user_id: 15,
  actions: [
    {
      type: 'view_product',
      product_id: 13,
      timestamp: ISODate('2026-03-13T13:11:42.098Z'),
      details: 'Viewed product page'
    },
    {
      type: 'add_to_cart',
      product_id: 31,
      timestamp: ISODate('2026-03-13T13:11:42.098Z'),
      details: 'Added product to cart'
    }
  ]
},
{
  _id: ObjectId('69b40d0e5fc2c40c588563d4'),
  user_id: 16,
  actions: [
    {
      type: 'view_product',
      product_id: 25,
      timestamp: ISODate('2026-03-13T13:11:42.098Z'),
      details: 'Viewed product page'
    },
```

```
    {
      type: 'add_to_cart',
      product_id: 6,
      timestamp: ISODate('2026-03-13T13:11:42.098Z'),
      details: 'Added product to cart'
    }
  ]
},
{
  _id: ObjectId('69b40d0e5fc2c40c588563d5'),
  user_id: 17,
  actions: [
    {
      type: 'view_product',
      product_id: 20,
      timestamp: ISODate('2026-03-13T13:11:42.099Z'),
      details: 'Viewed product page'
    },
    {
      type: 'add_to_cart',
      product_id: 34,
      timestamp: ISODate('2026-03-13T13:11:42.099Z'),
      details: 'Added product to cart'
    }
  ]
},
{
  _id: ObjectId('69b40d0e5fc2c40c588563d6'),
  user_id: 18,
  actions: [
    {
      type: 'view_product',
      product_id: 29,
      timestamp: ISODate('2026-03-13T13:11:42.100Z'),
      details: 'Viewed product page'
    },
    {
      type: 'add_to_cart',
      product_id: 33,
      timestamp: ISODate('2026-03-13T13:11:42.100Z'),
```

```
        details: 'Added product to cart'
      }
    ]
  },
  {
    _id: ObjectId('69b40d0e5fc2c40c588563d7'),
    user_id: 19,
    actions: [
      {
        type: 'view_product',
        product_id: 8,
        timestamp: ISODate('2026-03-13T13:11:42.101Z'),
        details: 'Viewed product page'
      },
      {
        type: 'add_to_cart',
        product_id: 49,
        timestamp: ISODate('2026-03-13T13:11:42.101Z'),
        details: 'Added product to cart'
      }
    ]
  }
]
Type "it" for more
ecommerce> it
[
  {
    _id: ObjectId('69b40d0e5fc2c40c588563d8'),
    user_id: 20,
    actions: [
      {
        type: 'view_product',
        product_id: 17,
        timestamp: ISODate('2026-03-13T13:11:42.102Z'),
        details: 'Viewed product page'
      },
      {
        type: 'add_to_cart',
        product_id: 30,
        timestamp: ISODate('2026-03-13T13:11:42.102Z'),
```

```
        details: 'Added product to cart'
      }
    ]
  }
]
```